Alain Pascal Francés

Hydrogeologist, PhD · Assistant Researcher, LNEG (Lisbon, Portugal)

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Profile

Hydrogeologist at LNEG, Portugal's National Laboratory of Energy and Geology, with more than 25 years of experience in Hydrogeology and Geoscientific Information Management. Combines numerical groundwater modelling (MODFLOW 6, FloPy, PEST) and coupled unsaturated–saturated zone modelling (MARMITES, of which he is the developer) with hydrogeophysics (ERT, TDEM/FDEM, MRS), remote sensing and GIS.

Currently involved in:

- The EU project DRYAD (https://dryad-project.eu/), from which he is the leader of Work Package 3 (modelling of Mediterranean agro-silvo-pastoral ecosystems)
- The update of the Hydrogeological Map of Mozambique (DNGRH/World Bank) Capacity building through PanAfGeo (https://panafgeo.eurogeosurveys.org/)

Position and key projects

since 2021 — **Assistant Researcher** | LNEG, Geology, Hydrogeology and Coastal Geology Unit
Hydrogeology and Geoscientific Information Management. At LNEG and its predecessor institutes (IGM, INETI) since 1999.

2024–present — **DRYAD – Leader of Work Package 3** | EU project
Nature-based solutions for the resilience of Mediterranean agro-silvo-pastoral ecosystems WP3: modelling of Mediterranean agro-silvo-pastoral ecosystems) dryad-project.eu

2024–present — **Updating and Upgrading the Hydrogeological Map of Mozambique** | World Bank, DNGRH
Contract DNGRH-346682-CS-QCBS with the National Directorate of Water Resources Management.

2024–present — **PanAfGeo** | EuroGeoSurveys
Trainer in geoinformation (GIS, databases) and hydrogeology for African geological surveys.

2015–2022 — **PLANAGEO – National Geology Plan of Angola** | Head of digital mapping and hydrogeology
Coordination of 18 geological sheets at 1:250 000 and 7 at 1:100 000; hydrogeological map at 1:1 000 000; hydrogeology of the Kalahari–Ohangwena (Cunene) aquifer system with TDEM and drilling.

2016–2019 — **Groundwater surveillance of the Neves-Corvo mine** | PVCA-NC
Groundwater flow model and reporting.

Education

2015 — **PhD in Hydrogeology** | University of Twente, ITC (the Netherlands)
Integration of hydrogeophysics and remote sensing with coupled hydrological models. FCT doctoral grant.

2008 — **MSc in Water Resources and Environmental Management** | ITC (the Netherlands)
Groundwater Assessment and Management; graduated with distinction (90/100).

1995 — **Maîtrise des Sciences de la Terre (Geology)** | Université Paul Sabatier, Toulouse
Final year at the University of Lisbon (Erasmus).

Expertise

Modelling	MODFLOW 6, MODFLOW-NWT (UZF, SFR, LAK), FloPy, PEST; MARMITES–MODFLOW; recharge and evapotranspiration partitioning; conceptual models; pumping tests
Geophysics	Geoelectrical (Syscal Pro, ABEM SAS4000, SuperSting R8); TDEM (TEM-FAST 48); FDEM (EM38/31/34); magnetic resonance sounding (NUMIS Lite)
Data & GIS	ArcGIS Pro, QGIS, SAGA GIS; PostgreSQL; Python, R; satellite image analysis; automatic monitoring networks

Supervision, peer review and service

Supervision	2 MSc theses ongoing (Universidade Eduardo Mondlane, Mozambique, co-supervision); 4 MSc theses at ITC (2009–2010); 2 undergraduate projects (2001–2002)
Reviewer	Hydrogeology Journal, Journal of Hydrology, Journal of Hydrology: Regional Studies, Journal of Applied Geophysics,

	Near Surface Geophysics, Geophysical Prospecting, Environmental Earth Sciences, Frontiers in Water, Pure and Applied Geophysics, and others
Membership	International Association of Hydrogeologists (IAH), since 2000
Languages	French (native), Portuguese (fluent), English (fluent), Spanish (spoken, reading)

Selected publications

Mendes, M. P., F. Salbitano, M. W. Lubczyński, A. J. Villanueva, A. Pantera, **A. P. Francés**, A. Silva and J. Samper (2026). The Role of Mediterranean Agrosilvopastoral Ecosystems in and Around Cities in Mitigating Climate Extremes. In: *Construction, Energy, Environment and Sustainability. Proceedings of CEES 2025, Vol. 3: Environment*. Vol. 745. Lecture Notes in Civil Engineering. Springer Nature Singapore, pp. 419–427. DOI: 10.1007/978-981-95-1818-0_42.

García-Lobón, J. L., E. Merino-Martínez, J. Fernández, J. Carvalho, M. J. Batista, J. Escuder, V. Lisboa, R. Martín-Banda, **A. P. Francés**, E. Ramalho, I. Martín-Méndez, C. Rey-Moral, M. d. C. Feria, T. Mochales, E. Ferreira, I. Cuervo, J. Manuel and A. Da Mata Lourenço Victorino (2025). The UTE-PLANAGEO Project (National Geology Plan of Angola). Objectives, methods, and results. *Boletín Geológico y Minero* 136.1, 001. DOI: 10.21701/bolgeomin/136.1/001.

Ramalho, E. C., **A. P. Francés**, R. Martín-Banda, J. M. Llorente, T. Mateus, F. Monteiro Santos, I. Cuervo, J. L. García-Lobón, V. Dala, M. Ditutala, A. Famorosa and A. Da Mata Lourenço Victorino (2025). Hydrogeological investigation in the Kalahari basin within the framework of the PLANAGEO Project. *Boletín Geológico y Minero* 136.2, 007. DOI: 10.21701/bolgeomin/136.2/007.

Francés, A. P., E. C. Ramalho, F. Monteiro Santos, J. M. Llorente, T. Mateus, R. Martín-Banda, I. Cuervo, J. L. García Lobón, V. Dala, M. Ditutala, A. Famorosa and A. da Mata Victorino (2024). Contribution of the time domain electromagnetic method to the study of the Kalahari transboundary multilayered aquifer systems in Southern Angola. *Hydrogeology Journal* 32.6, pp. 1709–1727. DOI: 10.1007/s10040-024-02822-x.

Francés, A. P. and M. W. Lubczynski (2023). Partitioning and sourcing of evapotranspiration using coupled MARMITES-MODFLOW model, La Mata catchment (Spain). *Frontiers in Water* 5. DOI: 10.3389/frwa.2023.1055934.

Ramalho, E. C., **A. P. Francés**, F. M. Santos and A. d. M. Victorino (2023). 3D electrical structure definition of aquifer systems in the Kalahari basin in Southern Angola based on legacy data reprocessing. *Journal of Applied Geophysics* 211, p. 104968. DOI: 10.1016/j.jappgeo.2023.104968.

Daoud, M. G., M. W. Lubczynski, Z. Vekerdy and **A. P. Francés** (2022). Application of a novel cascade-routing and reinfiltration concept with a Voronoi unstructured grid in MODFLOW 6, for an assessment of surface-water/groundwater interactions in a hard-rock catchment (Sardon, Spain). *Hydrogeology Journal* 30.3, pp. 899–925. DOI: 10.1007/s10040-021-02430-z.

Balugani, E., M. Lubczynski, L. Reyes-Acosta, C. van der Tol, **A. P. Francés** and K. Metselaar (2017). Groundwater and unsaturated zone evaporation and transpiration in a semi-arid open woodland. *Journal of Hydrology* 547, pp. 54–66. DOI: 10.1016/j.jhydrol.2017.01.042.

Francés, A. P., E. C. Ramalho, J. Fernandes, M. Groen, R. Hugman, M. A. Khalil, J. De Plaen and F. A. Monteiro Santos (2015). Contributions of hydrogeophysics to the hydrogeological conceptual model of the Albufeira-Ribeira de Quarteira coastal aquifer in Algarve, Portugal. *Hydrogeology Journal* 23.7, pp. 1553–1572. DOI: 10.1007/s10040-015-1282-x.

Baroncini-Turricchia, G., **A. P. Francés**, M. Lubczynski, J. Martínez-Fernández and J. Roy (2014). Integrating MRS data with hydrologic model - Carrizal Catchment (Spain). *Near Surface Geophysics* 12.2, pp. 255–269. DOI: 10.3997/1873-0604.2014003.

Francés, A. P., M. W. Lubczynski, J. Roy, F. A. Santos and M. R. Mahmoudzadeh Ardekani (2014). Hydrogeophysics and remote sensing for the design of hydrogeological conceptual models in hard rocks – Sardón catchment (Spain). *Journal of Applied Geophysics* 110, pp. 63–81. DOI: 10.1016/j.jappgeo.2014.08.015.

Francés, A. P. and M. W. Lubczynski (2011). Topsoil thickness prediction at the catchment scale by integration of invasive sampling, surface geophysics, remote sensing and statistical modeling. *Journal of Hydrology* 405.1-2, pp. 31–47. DOI: 10.1016/j.jhydrol.2011.05.006.

Full list of publications: orcid.org/0000-0002-1823-3094